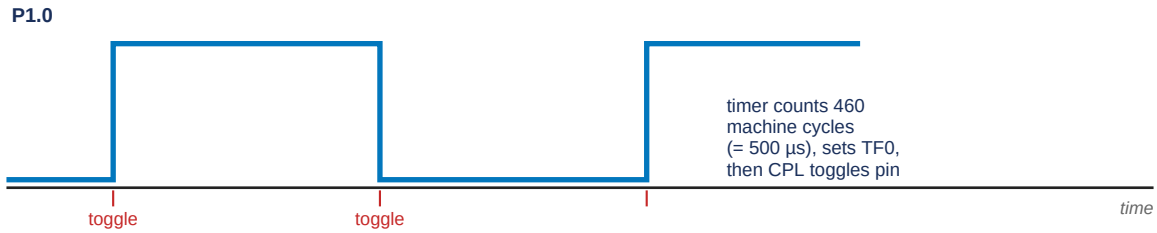
All 10 questions are ONE question in disguise: toggle a pin every half-period with a timer delay. Only 4 things change: **pin**, **timer**, **mode**, **reload values TH/TL**. Learn the method once → answer all 10.

A • The core idea (works for every question)



- **Square wave** = pin HIGH for half period, LOW for half period. So: **toggle the pin (CPL), wait half-period with the timer, repeat forever**.
- **Duty cycle** generalises it: HIGH for duty% × T, LOW for the rest — two different delays instead of one.
- **8051 timers count machine cycles**, not µs: with an 11.0592 MHz crystal one machine cycle = 12 ÷ 11.0592 MHz = **1.085 µs**. Timer overflow flag (TF0/TF1) sets automatically when the count rolls over.

TMOD (89H) — set ONCE at program start

GATE	C/T	M1	M0	GATE	C/T	M1	M0
TIMER 1				TIMER 0			

M1 M0 = 01 → Mode 1 (16-bit) • 10 → Mode 2 (8-bit auto-reload)

TCON (88H) — only 4 bits matter here

TR0	starts Timer 0
TF0	=1 when Timer 0 overflows
TR1	starts Timer 1
TF1	=1 when Timer 1 overflows

B • The 4-step recipe + formulas (write this skeleton in every answer)

```
        MOV  TMOD,#xxH ; STEP 1: select timer + mode (table below)
        MOV  THx,#__H   MOV TLx,#__H ; STEP 2: load reload value = 65536 – count (Mode 1)
        SETB TRx ; STEP 3: start the timer running
WAIT:  JNB  TFx,WAIT ; STEP 4: wait until overflow flag sets (count reached)
        CLR  TRx   CLR  TFx ; stop timer, clear flag → ready for next half-period
```

Quantity	Formula	Example (Q1: 1 kHz, 50%)
Period	**T = 1 / f**	T = 1/1000 = 1 ms
Half period	**t = T/2**	t = 500 µs
Duty split	**t_HIGH = duty% × T ; t_LOW = T – t_HIGH**	25%: 250 µs HIGH / 750 µs LOW
Timer count	**count = t / 1.085 µs**	500/1.085 = 460.8 ≈ 460
Mode 1 reload	**values = 65536 – count → THx:TLx**	65536–460 = FE34H
Mode 2 reload	**THx = 256 – count**	50 µs: 256–46 = D2H
TON / TOFF form	**T = TON + TOFF**	Q6: 1ms+1ms → 500 Hz square wave

C • Ready-made reload values (11.0592 MHz — memorise the bold ones)

Delay (Mode 1)	count	values = 65536–count	THx : TLx	Used in
250 µs	230	65306 = FF1AH	**FF : 1A**	Q3, Q4, Q5
500 µs	460	65076 = FE34H	**FE : 34**	Q1
1 ms	922	64614 = FC66H	**FC : 66**	Q2, Q6
5 ms	4608	60928 = EE00H	**EE : 00**	Q7
10 ms	9216	56320 = DC00H	**DC : 00**	Q9
max 71.1 ms	65536	0000H	00 : 00	limit!

Delay (Mode 2, 8-bit)	count	THx = 256–count	TMOD	Used in
50 µs	46	**210 = D2H**	**02H**	Q8
max 277.8 µs	256	00H	—	limit!

Mode 1 vs Mode 2: Mode 1 = 16-bit, up to 71.1 ms, must **reload TH/TL every time**. Mode 2 = 8-bit auto-reload (max 277.8 µs) → load THx once, never stop, just clear TF. **Longer delays (Q9: 0.5 s)** → run a 10 ms timer delay inside a counter loop (R2 = 50).

Q1 • 1 kHz, 50% duty @ P1.0, Timer 0 — THE TEMPLATE (learn perfectly, reuse everywhere)

Calculation: T = 1/1 kHz = 1 ms → half-period = 500 µs → count = 500/1.085 = 460.8 ≈ **460** → values = 65536 – 460 = 65076 = **FE34H** → **`TH0 = FE, TL0 = 34`** • Timer 0 Mode 1 → **`TMOD = 01H`**

```
        ORG  0000H ; reset address
        MOV  TMOD,#01H ; Timer 0, Mode 1 (16-bit)
AGAIN: CPL  P1.0 ; toggle output pin (0→1 or 1→0)
        MOV  TH0,#0FEH ; reload values FE34H (500 µs worth)
        MOV  TL0,#34H ; reload EVERY pass — Mode 1 forgets
        SETB TR0 ; start Timer 0
HERE:  JNB  TF0,HERE ; wait: TF0 sets after 460 counts = 500 µs
        CLR  TR0 ; stop timer
        CLR  TF0 ; clear overflow flag for the next half
        SJMP AGAIN ; toggle again → HIGH/LOW alternate → 1 kHz
        END
```

Why CPL works: the pin is LOW for 500 µs, toggled HIGH for 500 µs, toggled LOW … 50% duty comes free. **Q2 and Q3 are the SAME answer** — only three things change:

Q	f	T	half = t	count = t/1.085	values	Program changes
Q2	500 Hz	2 ms	1 ms	921.7 ≈ 922	64614 = FC66H	pin **`P1.1`**, **`TH0=FCH`**, **`TL0=66H`**
Q3	2 kHz	0.5 ms	250 µs	230.4 ≈ 230	65306 = FF1AH	pin **`P1.2`**, **`TH0=FFH`**, **`TL0=1AH`**

Q4 • 1 kHz, 25% duty @ P1.3, Timer 0 — duty cycle version

Calculation: T = 1 ms • **t_HIGH = 25% × 1 ms = 250 µs** → count 230 → **FF1AH** • **t_LOW = 750 µs** → count = 750/1.085 = 691.2 ≈ 691 → 65536–691 = 64845 = **FD4DH**

```
        ORG  0000H ; 1 kHz, HIGH 250 µs / LOW 750 µs
        MOV  TMOD,#01H ; Timer 0, Mode 1
LOOP:  SETB P1.3 ; → pin HIGH: ON-time begins
        ACALL D250 ; wait 250 µs (= 25% of 1 ms)
        CLR  P1.3 ; → pin LOW: OFF-time begins
        ACALL D750 ; wait 750 µs (= 75%)
        SJMP LOOP ; repeat → period stays 1 ms
D250:  MOV  TH0,#0FFH ; reload FF1AH (230 × 1.085 = 250 µs)
        MOV  TL0,#1AH
        SETB TR0
W1:    JNB  TF0,W1 ; wait for overflow
        CLR  TR0
        CLR  TF0
        RET
D750:  MOV  TH0,#0FDH ; reload FD4DH (691 × 1.085 ≈ 750 µs)
        MOV  TL0,#4DH
        SETB TR0
W2:    JNB  TF0,W2
        CLR  TR0
        CLR  TF0
        RET
        END
```

Q5 (75% duty, P1.4) = same program with the two delays swapped and pin changed: SETB **P1.4** → ACALL **D750**; CLR → ACALL **D250**. HIGH = 750 µs (FD4DH), LOW = 250 µs (FF1AH). Same period 1 ms.

Q6 • TON = 1 ms, TOFF = 1 ms @ P2.1, Timer 1

Calculation: **T = TON + TOFF = 2 ms** → this is a 500 Hz square wave • half-period = TON = 1 ms → count = 922 → **FC66H** • Timer 1 Mode 1 → **TMOD = 10H** (nibble moves!) • flags become TR1/TF1

```
        ORG  0000H ; TON = TOFF = 1 ms
        MOV  TMOD,#10H ; TIMER 1, Mode 1 (01H in the upper nibble)
AGAIN: CPL  P2.1 ; toggle output pin
        MOV  TH1,#0FCH ; reload FC66H = 922 counts = 1 ms
        MOV  TL1,#66H
        SETB TR1 ; start Timer 1 (not TR0!)
HERE:  JNB  TF1,HERE ; wait for Timer 1 overflow (1 ms)
        CLR  TR1
        CLR  TF1
        SJMP AGAIN ; repeat: 1 ms ON + 1 ms OFF
        END
```

Q7 • 100 Hz, 50% duty @ P2.2, Timer 1 in Mode 1

Calculation: T = 1/100 Hz = 10 ms → half = 5 ms = 5000 µs → count = 5000/1.085 = 4608.3 ≈ **4608** → values = 65536 – 4608 = 60928 = **EE00H** → **`TH1 = EE, TL1 = 00`** • Timer 1 Mode 1 → **`TMOD = 10H`**

```
        ORG  0000H ; 100 Hz → very long half-period (5 ms)
        MOV  TMOD,#10H ; Timer 1, Mode 1
AGAIN: CPL  P2.2 ; toggle output pin
        MOV  TH1,#0EEH ; reload EE00H = 4608 counts = 5 ms
        MOV  TL1,#00H
        SETB TR1 ; start Timer 1
HERE:  JNB  TF1,HERE ; wait 5 ms
        CLR  TR1
        CLR  TF1
        SJMP AGAIN ; repeat → 10 ms period = 100 Hz
        END
```

Q8 • 10 kHz, 50% duty @ P2.3, Timer 0 in Mode 2 (8-bit auto-reload)

Calculation: T = 1/10 kHz = 100 µs → half = 50 µs → count = 50/1.085 = 46.1 ≈ **46** → Mode 2: **`TH0 = 256 – 46 = 212 = D2H`** • **`TMOD = 02H`** • reload is AUTOMATIC → load once, no reload, no CLR TR0

```
        ORG  0000H ; 10 kHz → tiny 50 µs halves → Mode 2 is ideal
        MOV  TMOD,#02H ; Timer 0, Mode 2 (8-bit AUTO-RELOAD)
        MOV  TH0,#0D2H ; reload value loaded ONCE — kept in TH0
AGAIN: CPL  P2.3 ; toggle output pin
        MOV  TL0,#0D2H ; first count into TL0 (later automatic)
        SETB TR0 ; start Timer 0 — leave it running forever
HERE:  JNB  TF0,HERE ; wait 46 counts = 50 µs
        CLR  TF0 ; clear flag; TL0 auto-reloads from TH0
        SJMP AGAIN ; repeat → 100 µs period = 10 kHz
        END
```

Mode 2 exam points: (1) only 8 bits → max delay 256 × 1.085 = 277.8 µs — fine here (50 µs). (2) **THx keeps the reload value permanently**; on overflow TLx is reloaded by hardware. (3) no need for CLR TR0 or reloading TH/TL — that's exactly why Mode 2 suits fast waves. (4) first start: you must still load TL0 once in software.

Q9 • 1 Hz, 50% duty @ P3.0, Timer 0 — delay too long? nest it!

Calculation: T = 1 s → half = 0.5 s … but Mode 1 max = 65536 × 1.085 µs = **71.1 ms only!** → use 50 × 10 ms inside a loop: 10 ms → count 9216 → **DC00H** (**TH0=DCH, TL0=00H**), **R2 = 50** repetitions

```
        ORG  0000H ; 1 Hz: pin must toggle every 0.5 s
        MOV  TMOD,#01H ; Timer 0, Mode 1
        MOV  R2,#50 ; 50 × 10 ms = 0.5 s half-period
MAIN:  CPL  P3.0 ; toggle output pin
AGAIN: MOV  TH0,#0DCH ; reload DC00H = 9216 counts = 10 ms
        MOV  TL0,#00H ; reloaded EVERY 10 ms round
        SETB TR0
HERE:  JNB  TF0,HERE ; wait 10 ms
        CLR  TR0
        CLR  TF0
        DJNZ R2,AGAIN ; 0.5 s over? no → next 10 ms
        MOV  R2,#50 ; yes → restart the 50 count
        SJMP MAIN ; toggle → 1 Hz square wave
        END
```

Q10 • TON = 0.5 s, TOFF = 1 s @ P2.0 WITHOUT timers (software delay)

Calculation: only loops: **DJNZ takes 2 machine cycles = 2.17 µs** → inner loop R7=230 → 230 × 2.17 ≈ 0.5 ms → × R6=100 → 50 ms → × R5=10 → **0.5 s (ON)**; R5=20 → **1 s (OFF)** (three-level nesting, since 0.5 s ■ 65 ms limit of one 16-bit loop)

```
        ORG  0000H ; software delays only — TMOD/TRx/TFx unused
MAIN:  SETB P2.0 ; pin HIGH → ON time starts
        ACALL D500MS ; 0.5 s software delay
        CLR  P2.0 ; pin LOW → OFF time starts
        ACALL D1S ; 1 s software delay
        SJMP MAIN ; repeat → T = 1.5 s, 33% duty
D500MS:MOV  R5,#10 ; — 10 × 100 × 0.5 ms = 0.5 s —
L1:    MOV  R6,#100
L2:    MOV  R7,#230 ; 230 × 2 cycles × 1.085 µs ≈ 0.5 ms
L3:    DJNZ R7,L3 ; inner loop: 230 passes
        DJNZ R6,L2 ; middle loop: 100 ×
        DJNZ R5,L1 ; outer loop: 10 ×
        RET
D1S:   MOV  R5,#20 ; — 20 × 100 × 0.5 ms = 1 s —
M1:    MOV  R6,#100
M2:    MOV  R7,#230
M3:    DJNZ R7,M3
        DJNZ R6,M2
        DJNZ R5,M1
        RET
        END
```

Exam tips for all 10: ① always write the calculation line first (marks!) — T, half-period, count, values, TMOD • ② don't forget **CLR TFx** (Mode 1) after waiting • ③ Timer 1 ⇒ TMOD nibble **#10H**, TR1, TF1, TH1/TL1 • ④ duty ≠ 50% ⇒ two different delays, period unchanged • ⑤ Mode 2 = auto-reload: load THx once, no CLR TRx • ⑥ delays > 71.1 ms ⇒ nested counter loops • ⑦ quote approximations (≈) — counts are rounded to integers.